

TONGMIAO XU

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EDUCATION

- MS in Computer Science** Aug 2025 – May 2027(expected)
Urbana, IL
- University of Illinois Urbana-Champaign, GPA 4.0/4.0
 - Related courses: Mobile Robotics(**SLAM, kalman filter..**) A+; Database System; Deep Generative Model; Reinforcement Learning; Deep Learning for Computer Vision
- BS in Electrical Engineering, Mechanical Engineering (Double major)** Sep 2021 – Jun 2025
Xi'an, China
- Xi'an Jiaotong University, GPA: 3.90/4.0 in Honors Engineering Program
 - Related courses: Program Design Method; Computer Network Theory; Data Science; Artificial Intelligence
 - Study Abroad to University of California, Berkeley in Fall 2023
Berkeley, CA
 - Related Courses: Mechatronics Design; Introduction to Embedded Systems; Advanced Control Systems

RELATED EXPERIENCE

- University of Illinois** Champaign, IL
- Research Assistant, **Real-to-Sim** Robot Policy Evaluation (Advisor: Prof. Svetlana Lazebnik, Yunzhu Li)* Sep 2025 – Present
- Combined **3D Gaussian Splatting** (3DGS) of the ALOHA robot workspace with **Mujoco** for policy evaluation.
 - Implemented pipelines for reconstruction, rendering, and composite simulation, supporting robust policy deployment including ACT, Diffusion, and Pi0.5 VLA model..
 - Contributed open-source code in collaboration with **lerobot**, **HuggingFace** and universities like NUS, NYU.
 - Expanded lerobot's ALOHA support and sim-to-real evaluation capabilities with moving cameras.
- National University of Singapore** Singapore
- Research Intern in Robot Learning with Dr. David Hsu [academic poster](#)* Jul 2024 – Apr 2025
- Developed **object-centric imitation learning models** for robotic manipulation (Fetch robot).
 - Advanced **3D representation learning** with RLbench to improve generalization by 67.5% across views/objects.
 - Presented work as an **academic poster at Eastern European Machine Learning Summer School (EEML 2025)**.

- Xi'an Jiaotong University** Xi'an, China
- Research Assistant in Machine Learning with Dr. Xiang Li [published paper](#)* Nov 2022 – May 2024
- Applied **event cameras** and **PointNet++** for machine fault diagnosis with point cloud representations.
 - Built novel **geometric data structure** for event streams; achieved robust classification under variables.
 - Published at **2024 IEEE Global Reliability & PHM Conference**.

- University of California, Berkeley** Berkeley, CA
- Project Participant in Embedded & Cyber-Physical Systems Course* Fall 2023
- Awarded **Second Prize** in Mobile Robot Design Social Impact competition.
 - Programmed **embedded C++** systems on Raspberry Pi for Bluetooth-based motion control in Linux.
 - Designed and implemented **localization and control algorithms** with gyroscopes, motors, and color sensors.
 - Achieved adaptive path following, line tracking, obstacle avoidance using a **finite state machine**.

OTHER EXPERIENCE

- Siebel School of Computing & Data Science, University of Illinois** Champaign, IL
- Graduate Teaching Assistant for Data Science Discovery and Database System* Aug 2025 – Present
- Lead weekly Python labs and office hours for machine learning, database system for up to 60 students.
 - Collaborate with professors to design course materials and assignments, demonstrating strong communication and teamwork skills recognized by faculty.
 - Improve students' practical coding skills and contribute to curriculum quality and receive a full TA evaluation.

ACTIVITIES & LEADERSHIP

- Best Research Paper Award**, EEML 2025 – Sarajevo, Bosnia and Herzegovina Jul 2025
- Best Paper Prize**, IEEE 2024 Global Reliability & PHM Conference Sep 2024
- National University Intelligent **Robotics Contest 1st Prize**, Shaanxi, China Aug 2024
- Leader** of the Hsue-shen Tsien Academic Study Assistance Group Aug 2022 – Aug 2023

SKILLS

- Programming:** Python, C/C++, MATLAB, MySQL
- Frameworks & Tools:** ROS, PyTorch, TensorFlow, OpenCV, Solidworks, Gazebo, Rviz, Git, LaTeX
- Hardware:** Arduino, Raspberry Pi, Nvidia Jetson